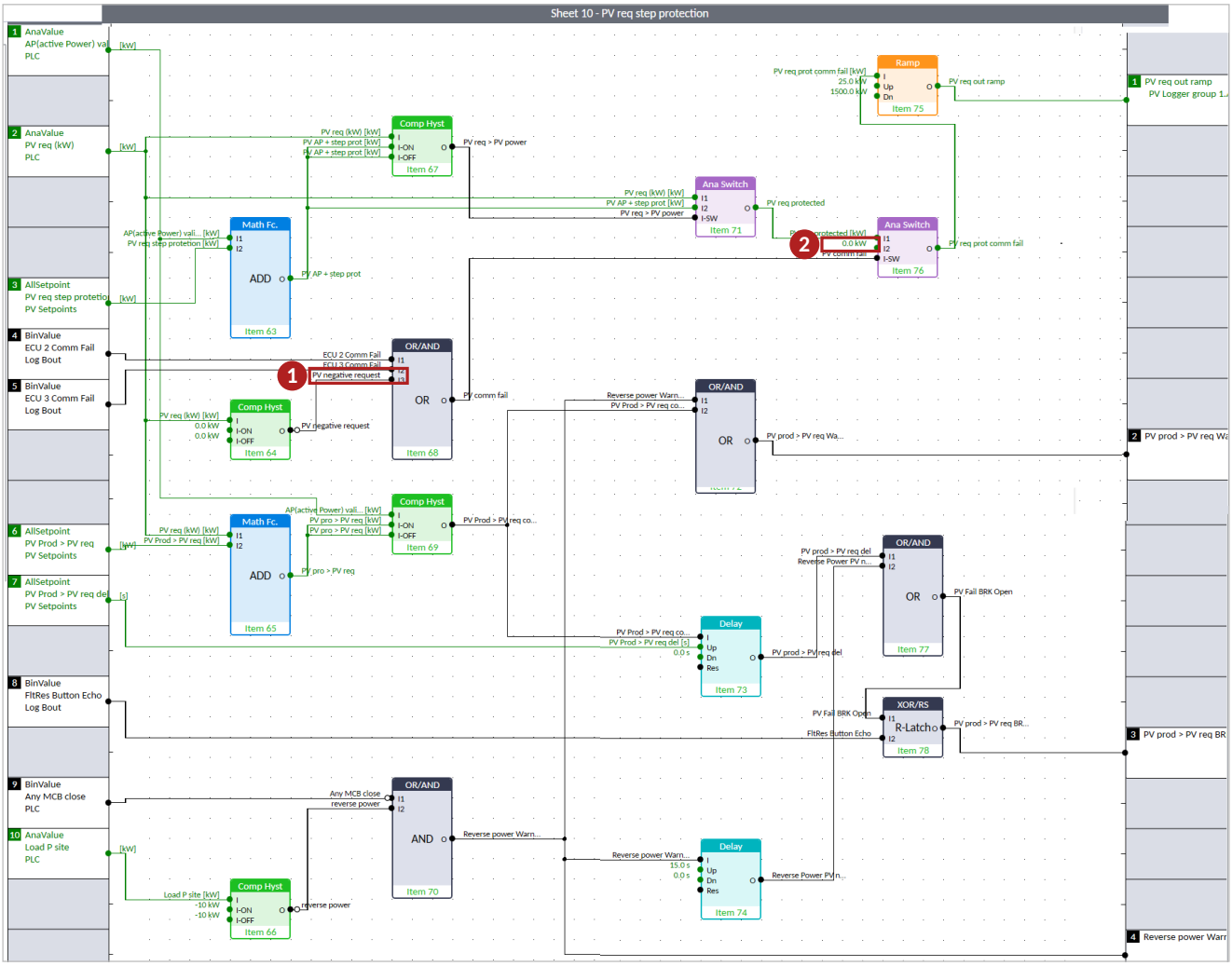
Sheet 10 today, and what goes wrong

Erbil plant - InteliNeo 5500 master - PLC sheet 10 "PV req step protection"



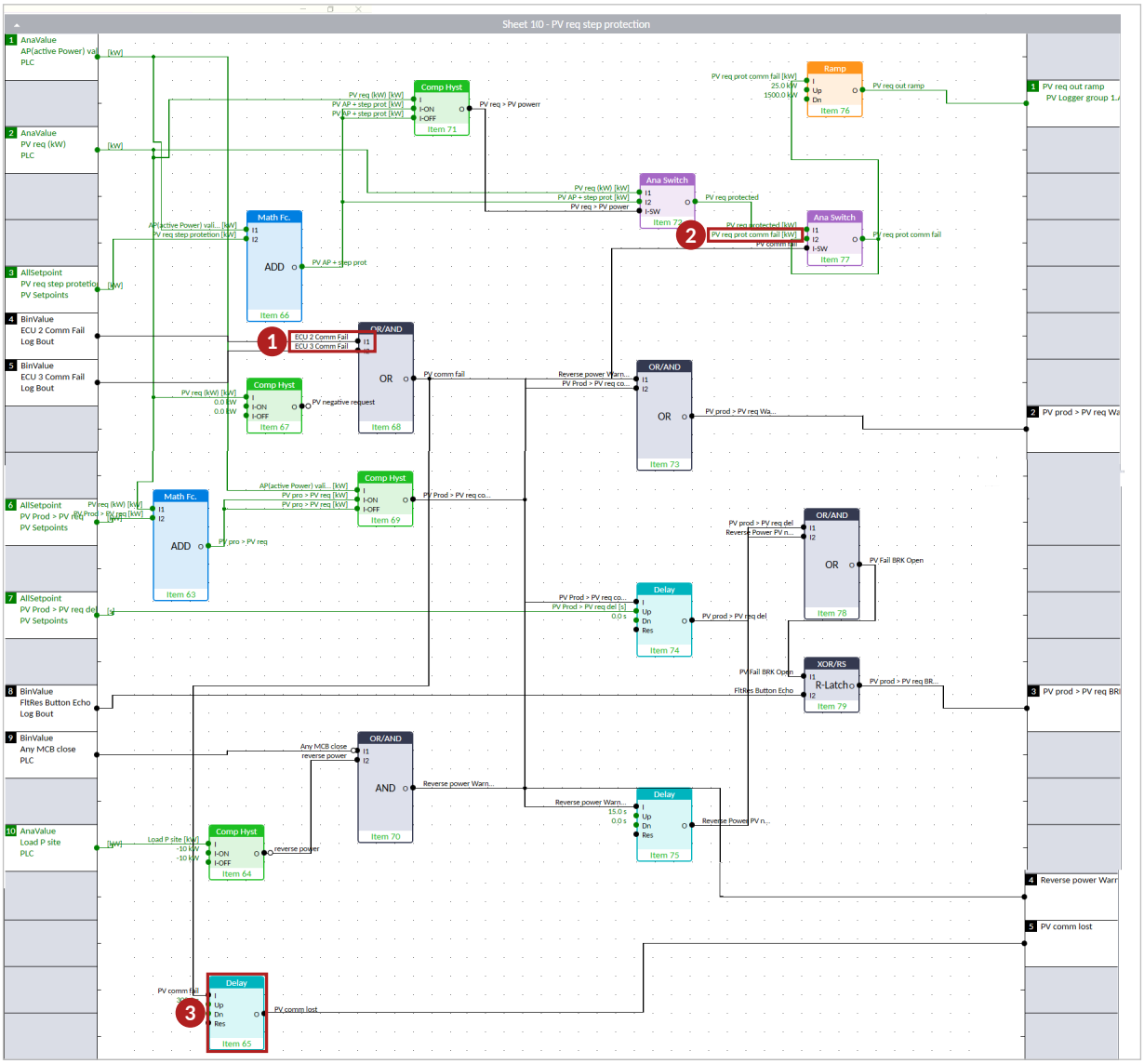
This sheet sends the solar plant its setpoint. Most of it is fine. The problem is one small switch, Item 76. Normally it just passes the request through. But if the controller misses even one reply from the PV logger, Item 68 (1) flips the switch, and the other side of the switch is a fixed 0.0 kW (2). So the solar plant gets told: zero.

The inverters do what they are told, within seconds. The gensets were running at 25 % and now they carry everything. The UPS logic sees that jump and sends its shedding command to the UPS breakers (whether the UPS really react is a separate question, the history shows breaker fail warnings). Then the reply comes back, and the solar has to climb up again, 100 kW at a time, about five minutes. This happens five to thirty times a day.

And here is the silly part: while the link is down, the zero never even reaches the logger. It arrives when the link is back, so it protects nothing. We just lose production every time.

What I want to change

Erbil plant - InteliNeo 5500 master - PLC sheet 10 "PV req step protection"



Three small edits, nothing else. The switch (now Item 77) gets its own output on its second input (2). So when a reply is missed, it just keeps sending the last setpoint. Nothing drops, nobody notices. Item 68 (1) now only listens to the two real comm fail signals, so if the controller really wants zero, zero still goes out. And one new timer, Item 65 (3), raises a flag called PV comm lost if the logger stays silent for five minutes. For now that flag only goes to an alarm.

Why I think we should do it

With a healthy link, the program does exactly what it does today. I ran the whole program block by block against our own history to check that. While the link is down, nothing is different from today either: the inverters keep their last setpoint in both versions, the gensets have 25 % in hand, the controller's own protections still work without the logger, and after five minutes we get the flag. If anyone wants it back, we load the archive from 14 October. One minute.

On the day of 14 October this change would have taken away three of the eleven UPS shedding commands and given us about 350 kWh more solar. And it is the first step before we can run fewer gensets, which is where the real fuel saving is.